

Using Hierarchical Models to Estimate Heterogeneous Effects

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Abstract

Conditional theories are ubiquitous in political science, but standard interaction models cannot handle this proliferation of theoretically informed modifiers. This paper describes why, when, and how to use hierarchical models to encompass multiple sources of heterogeneity and regularize estimates. Doing so is simple, and starts with identifying the key sources of heterogeneity and grouping observations accordingly. Next, fit a hierarchical model with two linked regressions, one connecting treatment with the outcome, and another that models the treatment effects with potential sources of heterogeneity and partially pools group estimates. This captures systematic and random variation in heterogeneous effects, encompasses many interactions, and fits commonly used modeling frameworks, but regularization in this model breaks down if there are many small groups. I document these claims with a simulation analysis and reanalysis of a published study.

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Social scientists often propose conditional theories (Clark and Golder, 2023) and are interested in how different units respond to the same stimulus for theoretical, normative or policy reasons. While most theories consider a single modifier, many theories propose distinct modifiers for the same independent variable. Therefore, multiple modifiers are common. For example, international relations experiments on audience costs (Tomz, 2007) have considered how foreign policy dispositions (Kertzer and Brutger, 2016), partisanship (Levendusky and Horowitz, 2012), gender (Barnhart et al., 2020; Schwartz and Blair, 2020) and policy preferences (Chaudoin, 2014) modify reactions to a leader backing down from a threat. Other audience cost experiments manipulate casualties (Davies and Johns, 2013), how the leader backs down (Lin-Greenberg, 2019), other elite reactions (Levendusky and Horowitz, 2012) and promises vs threats (Levy et al., 2015).

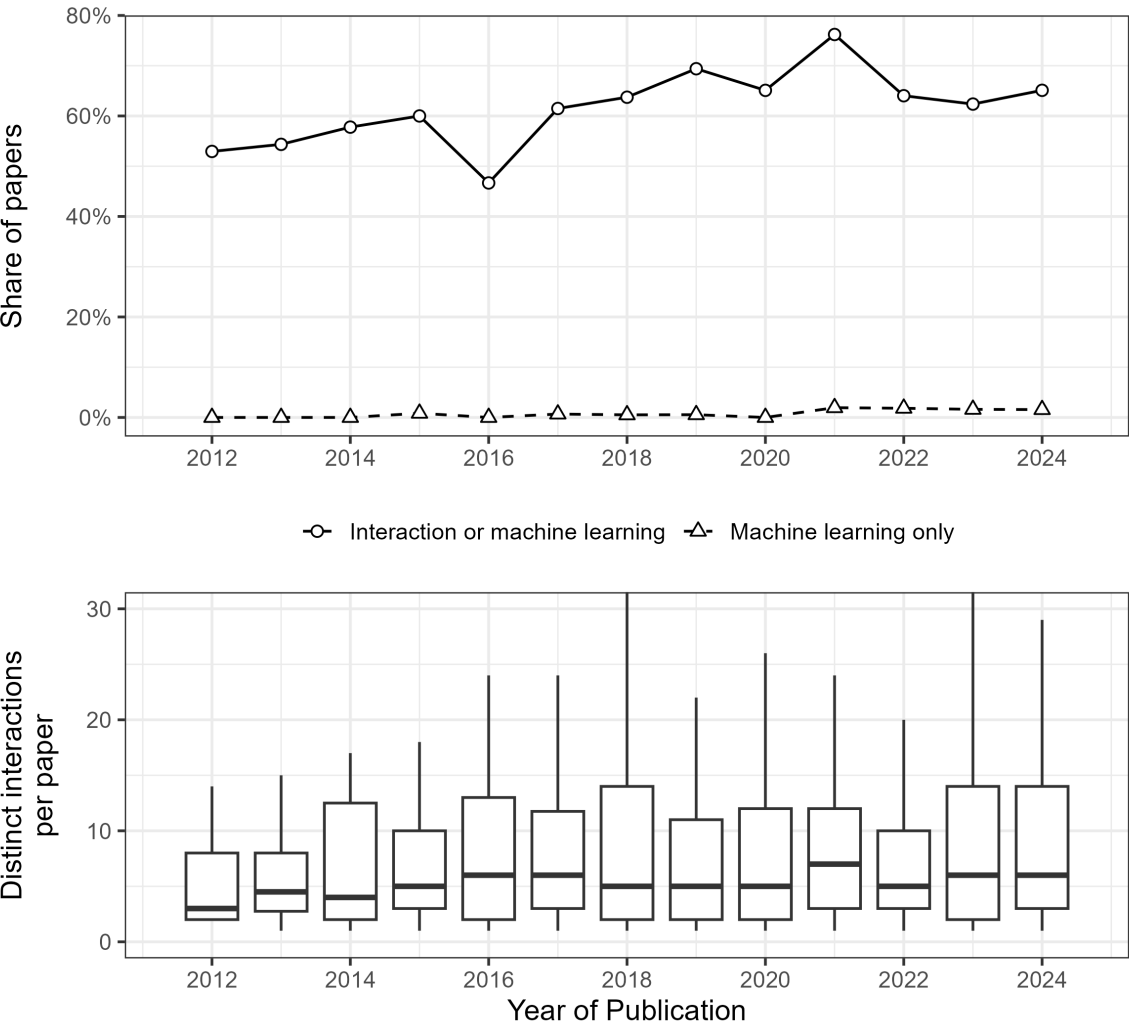
Many potential modifiers are evident in how scholars conduct quantitative research. In the *American Political Science Review*, *American Journal of Political Science* and *Journal of Politics*, many researchers estimate some modifier of treatment effects, either to test their argument or explore others. Figure 1 documents the prevalence of heterogeneous effects estimates from interactions or machine learning in Harvard Dataverse replication data from the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* from 2012 to 2024. In this figure I count the number of papers that fit at least one interaction or machine-learning model of effect heterogeneity in their replication data to show the prevalence of these tools. I did this by downloading all quantitative replication materials hosted in the Harvard dataverse, and then measuring the prevalence of heterogeneous effects estimates. I then used an open-weights LLM (gpt-oss-120b) to parse the code and identify models and interactions.¹

Interactions are very common. In most years, 60% of papers in the APSR, AJPS, and JOP estimate at least one interaction. Use of machine-learning tools for heterogeneous effects is much less common, and is usually in the single-digits.²

¹Blind hand validation of a stratified sample puts precision above 90%, with details in the appendix.

²As another indicator of interaction popularity, Brambor, Clark and Golder (2006) has 8,279 citations as of

Heterogeneous Effect Estimation in Leading Journals
2012-2024



Lower panel covers papers estimating at least one interaction and drops outliers from view

Figure 1. *Prevalence of heterogeneous effect estimates, 2012 to 2024.*

Moreover, researchers fit many interactions, as the median paper estimates 6 distinct interaction terms. A few outliers estimate many more. Most applied research relies on parametric interactions, with different theories and robustness checks leading to multiple interactions in most papers.

Unfortunately, the workhorse regression interactions that many scholars rely on are ill-suited to test the growing number of conditional arguments in social science. Scholars cannot ignore heterogeneity, but the most common tools either increase the risk of spurious results or are hard to interpret and use. Interaction terms and subgroup analysis are ubiquitous because they are relatively easy to interpret, but they increase the risk of spurious results. Many political science analyses have low power even to detect effects without interactions (Arel-Bundock et al., 2026). Adequate power to estimate even a single interaction can require significantly more data (Gelman, 2018), which may be prohibitively expensive or impossible. As a result, statistically significant heterogeneous effect estimates may be far too large—the result of noise in the data, not systematic differences. This problem is partially responsible for widespread issues replicating findings based on interactions (Simmons, Nelson and Simonsohn, 2011). Adding more modifiers makes interpretation harder and increases these power problems. Researchers might instead fit separate pairwise interactions for each modifier, but doing so multiplies models and corresponding multiple-testing problems.

Given multiple sources of heterogeneity, machine-learning tools such as random forests (Green and Kern, 2012; Wager and Athey, 2018), support vector machines (Imai and Ratkovic, 2013), and ensemble methods (Grimmer, Messing and Westwood, 2017; Künzel et al., 2019; Dorie et al., 2022) are more likely to avoid over-fitting. These machine learning algorithms usually have some regularization component and can discover complex patterns and high-dimensional variation across multiple modifiers.³ These tools can be difficult to interpret and

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³LASSO models sometimes address similar issues (Ratkovic and Tingley, 2017). Blackwell and Olson (2022) describe a lasso approach to interactions of a single modifier and all the variables in a model that sits between

implement, however, especially in smaller social science datasets. The relative lack of theoretical guidance and interpretability is especially problematic for testing the relative weight of multiple conditional arguments. If researchers want to compare the impact of different modifiers, machine learning models use quantities that many applied researchers are less familiar with, so these techniques are far less popular than interactions.

Despite a large statistics literature documenting how regression-based interpretation and regularization make Bayesian hierarchical models a useful tool for heterogeneous effects estimation (e.g., Feller and Gelman (2015); McElreath (2016); Dorie et al. (2022)), political scientists rarely use this tool. For instance, of the three applied political science citations of Feller and Gelman (2015), only Marquardt (2022) models treatment effects. The MAIHDA (Multilevel Analysis of Individual Heterogeneity and Discriminatory Accuracy) framework in intersectionality research is a noteworthy exception (e.g. Much, 2025). These researchers estimate distinct effects for groups defined by class, race and gender, and partial pooling helps give reasonable estimates for small subgroups in survey data. I use a similar logic to estimate how treatment effects vary across groups.

This paper provides practical guidance by explaining how, when and why to use hierarchical models of heterogeneous effects. The hierarchical varying slopes strategy I advocate here addresses the power shortcomings of interactions while allowing more theoretical guidance and comparison than machine learning. It does so by regularizing estimates of multiple modifiers, thus trading some bias of specific effects for reduced variance in effect estimates. Taking this side of the bias-variance tradeoff improves accuracy overall, and reduces errors about the sign and magnitude of non-zero effects.

Using hierarchical models is more flexible than standard interactions but easier to implement and interpret than machine learning. It preserves a straightforward structure while ac-

machine learning and linear regressions. This considers a single interaction of interest and bias from other unmodeled interactions with controls.

commodating more factors and ameliorating the downsides of subgroup analysis. The main weaknesses are that unlike machine learning, the hierarchical approach lacks the flexibility to discover high-dimensional heterogeneity and regularization will break down with many small groups. It can also be slow to fit on large datasets, depending on the underlying sampler.⁴

There are two key steps when theory and data make using hierarchical models worthwhile. First, researchers should identify potential modifiers of a treatment effect and use them to group observations and model treatment effects. Second, they should take that model of treatment effects and connect it to a model linking the treatment and outcome, implementing a type of varying slopes model. Modeling heterogeneous effects produces interpretable results, which helps researchers compare different modifiers. It also allows researchers to compare different sources of heterogeneous effects and describe effect variation. These are crucial advantages given many conditional theories.⁵

While frequentist estimation of hierarchical models is possible, Bayesian estimation is easy, more informative and more robust (Stegmueller, 2013). Bayesian estimation provides crucial information by connecting parameters through common prior distributions, thereby regularizing estimates and propagating uncertainty. Working with posterior distributions also gives researchers more flexibility to describe how and when effects vary and use prior predictive checks to examine the assumptions in their models. While computation and coding were once a barrier to employing Bayesian methods, fitting a wide range of hierarchical models is straightforward with the `brms` package in R (Bürkner, 2017).⁶

In the remainder of this paper, I start by describing how and when to estimate hierarchical models of heterogeneous effects. I then use simulations to compare OLS and hierarchical

⁴Goplerud (2021) considers a model that uses Bayesian structured sparsity to estimate which group coefficients are similar and which are different. In this approach, researchers use theory to inform potential groups, but the data determines common estimates for groups.

⁵The downside is that when partial pooling moves subgroup estimates towards the overall mean, testing claims about specific groups is not wise.

⁶I provide example code below and in the appendix.

estimates of treatment effects under different conditions. Finally, I demonstrate the process by analyzing a study by Barber and Pope (2019), who attempt to disentangle partisanship and ideology. The reanalysis documents the importance of jointly estimating modifiers in a single regularized model.

1 Hierarchical Modeling of Heterogeneous Effects

There are two steps in hierarchical models of heterogeneous effects with varying intercepts, varying slopes and group-level predictors of the slopes (Gelman and Hill, 2007). First, researchers must identify potential sources of heterogeneity, and think about the right model of heterogeneity. This will also depend on what variation is most important and interesting. Theory, policy concerns, or normative factors are all possible motivations.

This first step determines what heterogeneous effects a researcher estimates. It is analogous to researchers thinking through a regression specification and requires similar care. Researchers need to define what variation is most important, link heterogeneous effects to theory, and structure modeling. This is especially important for pre-registration, and could reduce the amount of exploratory analyses in registrations. Not thinking carefully about sources of heterogeneity will obfuscate results and can hinder model fitting.

There are three general approaches to defining key modifiers. First, researchers can use combinations of other treatments, especially when an intervention has several dimensions but theory emphasizes one of them. The experimental design determines modifiers, and the model estimates heterogeneous treatment effects. If researchers want to know how different issues shape the impact of elite foreign policy cues (Guisinger and Saunders, 2017), they could include indicators of issues, for instance. A similar application of hierarchical estimators for topic-sampling experiments estimates how a treatment effect varies across topics (Clifford and Rainey, 2024).

At the moment, researchers often estimate the impact of different treatment combinations with fully crossed interactions. This estimates the impact of a treatment across experimental strata, but risks spurious results by functionally estimating subgroup results. These estimates are often noisy, however, and can rely on 50 or fewer data points in smaller samples.

A second approach uses unit, demographic and contextual factors to estimate effect heterogeneity. Here, researchers examine what factors within or around units shape their response to an independent variable. Researchers could use a mix of individual and contextual factors to predict divergent consequences of a survey experiment treatment. Such a model might include factors such as an indicator of state of residence, age, gender, and race.⁷ For example, Alley (2021) uses alliance characteristics to examine when alliance membership increases or decreases military spending.

Third, researchers might use hierarchical models to address specific policy concerns. Policy analysts often want to know how an intervention impacts a specific population. Researchers might want to know if a job-training program improves employment outcomes for black women in the South, for instance. To do this, a researcher might specify a heterogeneous effects model with race, gender and region, plus additional controls or other factors.

After defining moderators and how they relate to treatment effects, the second step is fitting a hierarchical model that links a model of the outcome with a model of heterogeneity. Essentially, researchers model the outcome and the process that produces heterogeneous treatments. The model employs two connected regressions. One regression deals with the outcome, and includes the treatment effects.⁸

The other regression models the treatment effects. In this regression, theory should inform which modifiers predict the treatment effect. To regularize the estimates, this second equation

⁷Extrapolations to a representative sample of a subpopulation might require poststratification.

⁸If other units such as states define the groups, rather than combinations of modifying variables, then adding group-level predictors to this equation is essential. For example, in a model where an effect varies by state, adding state-level variables like ideology, population and GDP would avoid partially pooling small groups too far towards the overall mean.

must include varying intercepts for groups within the data. Groups can be based on combinations of discrete modifiers and binned continuous modifiers. As a result, this model includes both systematic and random variation in the impact of a key independent variable on different groups.

To maximize variation and preserve regularization, researchers should use continuous modifiers directly in the regression and bin them for setting groups. Not binning modifiers will create many small groups, leading the hierarchical component to break down. In the appendix, I document some additional best practices for continuous modifiers, including avoiding smooth priors for ordered effects across bins and the importance of bin boundaries relative to the number of bins.

Setting groups is a critical task that requires balancing detail against regularization. Larger groups can deviate more from the overall mean, as their data contributes more to the posterior than the prior. A few small groups means less need for a hierarchical model. Smaller groups will be shrunk more towards the overall mean. Too many small groups and the regularization component will have less information to work with. Using model fit criteria such as WAIC or leave-one-out cross validation can help guide group formulation. Pre-registration is also crucial, because this is the area with the greatest researcher degrees of freedom.

Should the variables in the heterogeneity equation also appear in the outcome equation? Here, the guidance of Brambor, Clark and Golder (2006) to include constituent terms in an interaction provides a helpful heuristic. If a modifier is correlated with treatment and the outcome, then it is also a useful control, as it has an independent relationship with the dependent variable in addition to modifying the treatment effects. Moreover, if untreated units in every group share the same intercept, the treatment effects absorb baseline outcomes. Including modifiers or group intercepts α_g in the outcome equation prevents this. The functional form of the two equations may differ, however.

I now briefly describe a generic hierarchical model.⁹ For ease of exposition, consider making between-unit comparisons based on an experimental treatment. Start with N units indexed by i , some of which receive a binary treatment T . Assume that the outcome variable y is normally distributed with mean μ_i and standard deviation σ .¹⁰ g indexes researcher-defined groups based on combinations of the modifying variables.

The outcome for each unit depends on a set of group intercepts α_g , an optional matrix of control variables $\mathbf{X}$, and a set of group treatment effects λ_g . The group intercepts are essential: they hold the baseline level of the outcome in each group, so that λ_g captures the effect of the treatment within groups rather than differences between them. When T is binary, estimated λ parameters for untreated units have no impact on the outcome and drop out of the likelihood. For a continuous treatment, effects will depend on the product of T_i and λ .

$$y_i \sim N(\mu_i, \sigma) \quad (\text{Likelihood})$$

$$\mu_i = \alpha_g + \lambda_g T + \mathbf{X}\beta \quad (\text{Outcome Equation})$$

$$\lambda_g = \theta_g + \mathbf{Z}\gamma \quad (\text{Heterogeneity Equation})$$

$$\theta_g \sim N(\mu_\theta, \sigma_\theta) \quad (\text{Group Varying Intercepts})$$

The heterogeneity equation then models those group treatment effects with a systematic and random component. The systematic component is a matrix of predictors $\mathbf{Z}$ and associated parameters γ . $\mathbf{Z}$ can mirror any regression specification researchers might use for an outcome; linear combinations of variables, interactions, smooth functions. Interactions of the modifiers could capture processes where combinations of modifiers produce non-additive jumps in heterogeneity, for instance.

⁹I split the equations for ease of exposition, but it is straightforward to see that this approach places priors on an additive interaction model of the form $y = T * (z_1 + z_2 + z_3 \dots) + \mathbf{X}\beta$

¹⁰Researchers can and should use binary, categorical and other outcome likelihoods.

The random component of the heterogeneity equation is a series of group-specific varying intercepts θ_g . These are critical, because they capture group deviations from the systematic trends expressed in the design matrix $\mathbf{Z}$. Group outliers in the λ estimates are partially pooled back towards the overall mean μ_θ . The variance parameter σ_θ controls the dispersion of the group effects around the overall mean.¹¹

The above model can be fit with Bayesian or frequentist methods, but Bayesian estimation offers important advantages. First, it is more flexible, and including prior information can facilitate model fitting. Priors on the α , β , and γ parameters will matter more in very small datasets. Priors on β should be relatively weak with observational data, because shrinking coefficients on confounders too much might reintroduce omitted variable bias (Hahn et al., 2018). The more important prior is for σ_θ , and depending on beliefs about potential dispersion across groups, half-normal or half student-t priors are common choices. This prior is what helps regularize estimates by pulling extreme groups towards the overall mean μ_θ .

Bayesian modeling offers additional information and a wealth of model-checking tools. STAN and brms, the software I provide guidance for here, facilitates prior predictive checks, where researchers simulate from the model to check what estimates are plausible with specific priors. Working with posterior distributions then provides a wealth of information about effect heterogeneity and propagates uncertainty. Last, Bayesian models can provide useful diagnostics such as divergent transitions that can be warnings of specification problems. It is often better to modify a model with fitting problems instead of brute-forcing it through the sampler.

Depending on the problem, researchers can extend this initial model in many ways. The

¹¹In brms, using non-linear syntax can express a model with a treatment, two controls, and three modifiers as: $y \sim \text{lambda} * \text{treat} + \text{controls}$, $\text{lambda} \sim \text{mod1} + \text{mod2} + \text{mod3} + (1 \mid \text{mod1}:\text{mod2}:\text{mod3})$, $\text{controls} \sim \text{mod1} + \text{mod2} + \text{mod3} + \text{control1} + \text{control2} + (1 \mid \text{mod1}:\text{mod2}:\text{mod3})$, $\text{nl} = \text{TRUE}$. The modifiers and the group intercept must appear in the controls equation as well, because together they form α_g . Omitting them imposes a common baseline on every group, and lambda then absorbs differences in the level of the outcome across groups rather than the effect of the treatment within them.

generative core of Bayesian hierarchical models adds significant flexibility. For instance, they might model the variance of the treatment effects or the outcome. Fitting splines or smooth functions is also possible, or researchers might extend the model in continuous group space with Gaussian process regression (Prati et al., 2023).

In interpreting these estimates, researchers should leverage the full range of information from the different parameters. First, the λ posteriors give the impact of the treatment on each group, and are the core quantity of interest. All λ s reflect a systematic component from the predictors in $\mathbf{Z}\gamma$ and random variation from θ . γ parameters can, depending on the regression, be interpreted as the impact of a change in a modifier on the treatment effects. For example, a γ of .1 on a binary modifier means that λ is .1 higher in expectation when the modifier is one, and .1 lower when it is zero. σ_θ thus measures treatment variation outside the systematic regression. Other techniques such as interactions in OLS with robust standard errors provide less information.

2 Performance on Simulated Data

To assess how this hierarchical model compares to interactions, I first assess their performance on simulated data. For ease of exposition, I consider machine-learning models in the appendix. The appendix compares LASSO, causal forest and BART, estimating the same group effects, and finds that the hierarchical model is usually more accurate.

In the simulations, I fix the number of observations, and then generate different scenarios with distinct groups and distributions of true effects across those groups. All of these scenarios give a group-specific effect of a binary treatment variable. I then estimate those effects with three models. First, ordinary least squares where the treatment interacts with an additive combination of the group variables. Second, ordinary least squares with fully crossed interactions between the grouping variables. The fully crossed model is somewhat extreme, but it appears

regularly in applied research, especially for subgroup effects in factorial experiments.¹² Third, a hierarchical model where an additive combination of grouping variables predicts treatment effects in each group and partial pooling regularizes effect variance. In the hierarchical model, each grouping variable enters the heterogeneity equation additively, with a random intercept for each group. Fully crossing the group variables would saturate the model and make the random intercepts redundant.

There are eight data-generating processes. One uses equal group sizes and normal effect distribution, which matches the partial pooling process well. Another uses heavy tails, drawing effects from a t-distribution with three degrees of freedom. A second creates imbalanced groups with the normal effect distribution, because larger groups have more information and can thus diverge more from the overall mean, while partial pooling shrinks smaller groups closer to the overall mean. To further test this mechanism, I implement another process where larger groups have larger effects, so partial pooling takes small groups closer to a potentially unrepresentative mean. Other tests include a sparse effect distribution with only a few different estimates, a bimodal distribution, and one with systematic effect variation that might be especially friendly to the additive interactions. The last process makes effects exactly additive in the modifiers, so the additive interaction model is correctly specified.

When I divide the 2,000 observations into groups based on combinations of three to seven binary variables and give equal probability of assigning observations to a group, there are between 8 and 128 unique groups. This gives approximately 250 observations per group at the low end and 16 observations at the high end. The imbalanced simulation can have hundreds of data points in one group and as few as two in another, but I fix the data-generating process so that all groups populate. Each condition uses 200 Monte Carlo replications. For ease of computation, accuracy results come from restricted maximum likelihood fits of the hierarchical

¹²An application in the appendix by Tomz and Weeks (2021) takes this approach.

model and error rates from Bayesian estimation.¹³

I then compare the models on root-mean squared error (RMSE) and the rate of type M and type S errors. The root-mean squared error of the coefficient estimates compared to the true value is the most general metric, and is the best because the hierarchical model's shrinkage will introduce bias but compensate with lower variance (Efron and Morris, 1975; Clifford and Rainey, 2024). The bias-variance trade-off is the core of the hierarchical advantage. Lower variance may overcome greater bias in specific coefficients and make the hierarchical model more accurate on average. Pooling to the treatment mean is conservative in some ways, but that conservative approach is useful when we never know the true distribution of treatment effects. RMSE also captures instances where an estimator exaggerates the magnitude of the coefficient.

I summarize the RMSE of the hierarchical and OLS models in Figure 2. All three models perform worse as the number of groups increases and each group becomes smaller and noisier. OLS with fully crossed interactions of grouping variables performs comparably to the hierarchical model when there are 8 groups, but RMSE increases dramatically as the number of groups increases. Unequal group sizes make this problem especially acute.

The additive OLS specification does not decay in the same way. For some group configurations, there is little difference as group size increases. The downside is consistently larger RMSE relative to the hierarchical model across every group size, except for when the effects are exactly additive. Under additive heterogeneity, a correctly specified additive interaction model is 3% more accurate.

The hierarchical model is usually best in RMSE, especially as groups proliferate. With few groups, some OLS models perform marginally better, but not clearly so. Like most models, accuracy degrades as groups proliferate, because there is greater scope for bias.

These patterns are the result of reduced variance overcoming greater bias in the hierarchical

¹³In the appendix, I document that variance hyper-priors do not change the results.

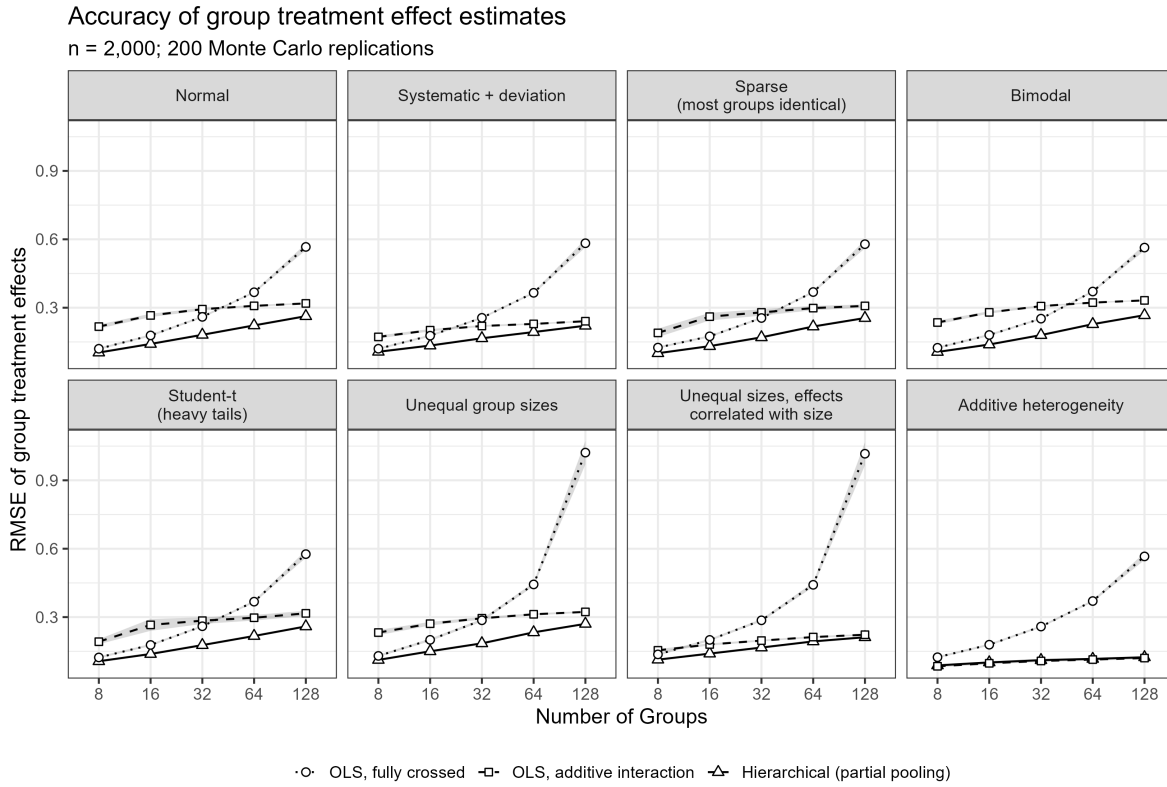


Figure 2. Comparison of OLS and hierarchical root mean squared error in group coefficient estimates. Simulation fixes the sample size at 2,000 observations and varies the number of groups. Bands give 95% Monte Carlo intervals.

estimates. I illustrate this pattern in Figure 3, which draws from the simulations with 8 and 128 groups with a normal effect distribution for ease of presentation. This figure plots the true against estimated values in those simulations for the three model specifications, using the normal distribution of true effects to illustrate. To clarify the pattern, I regress the true values on the estimated values.

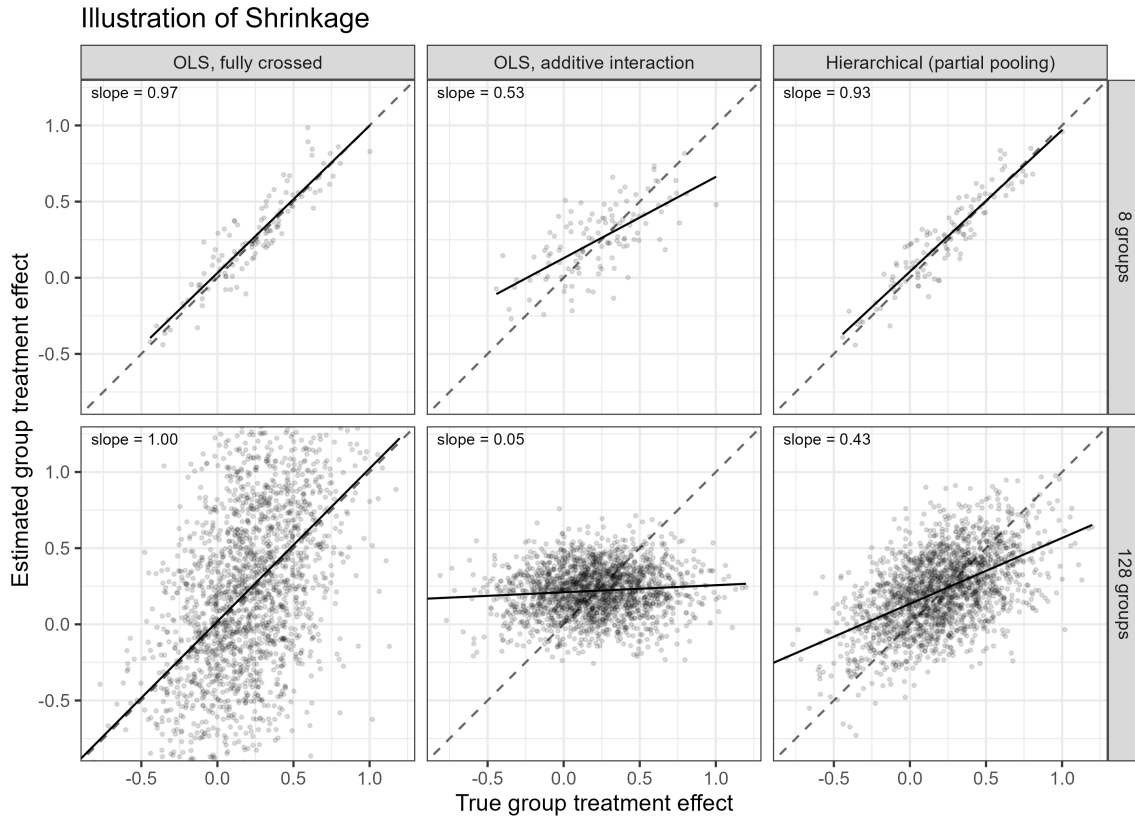


Figure 3. True and estimated effects from simulations with 8 and 128 groups. The black line regresses the true value on the estimates. Points on the dashed line are unbiased.

Under 8 groups, the hierarchical models and fully crossed OLS have similar slopes, with marginally less variance around the true values for the hierarchical model. This is why these two specifications have similar RMSE with a few groups. OLS with an additive interaction of treatment and modifiers gives a single common answer, smoothing out too much of the true variation.

At the other extreme of 128 groups, fully crossed interactions have significant noise around the true value. This variance increases RMSE, despite less bias in the overall trend. The additive model flattens out the overall trend, giving a single trend. It also has noise, but far less than the fully-crossed OLS, so the RMSE is much better. The hierarchical model falls in between. It has more bias than fully crossed OLS, but much lower variance. It recovers the true values better than the additive OLS specification, with somewhat more variation.

This is the bias-variance tradeoff in action. Hierarchical models are more biased than some specifications, but the reduction in variance makes them more accurate on average. This conservative approach is especially important when there are many modifiers and group size decreases.

Conservative estimation of subgroup effects is especially important when researchers use statistical significance thresholds. Hierarchical modeling reduces the risk that statistically significant results have the wrong sign or an exaggerated size (Gelman and Carlin, 2014). These Type S (sign) and Type M (magnitude) errors are just as important as whether an uncertainty interval includes zero. Misstating the direction of an effect misleads in theory testing, and magnitude errors fail to capture the substantive importance of an effect.

Figure 4 shows the rates of Type S and M errors in estimates where the 95% confidence or credible interval excludes zero.¹⁴ This figure plots error rates at 8, 32, and 128 groups. The rate gives the percentage of estimates with a sign error and the exaggeration ratio for Type M errors.

With sign errors, hierarchical models and fully crossed OLS perform similarly. The additive OLS model makes more sign errors because it fails to pick up adequate variation, especially as the number of groups increases. In some cases, as many as 1/3 of statistically significant effects are the wrong sign using additive OLS interactions. Other models make errors too, but they are far more rare.

¹⁴Some OLS estimates drop due to perfect collinearity and are excluded.

Errors of inference among subgroup effects declared significant

Intervals excluding zero: posterior for the hierarchical model, delta method for OLS.

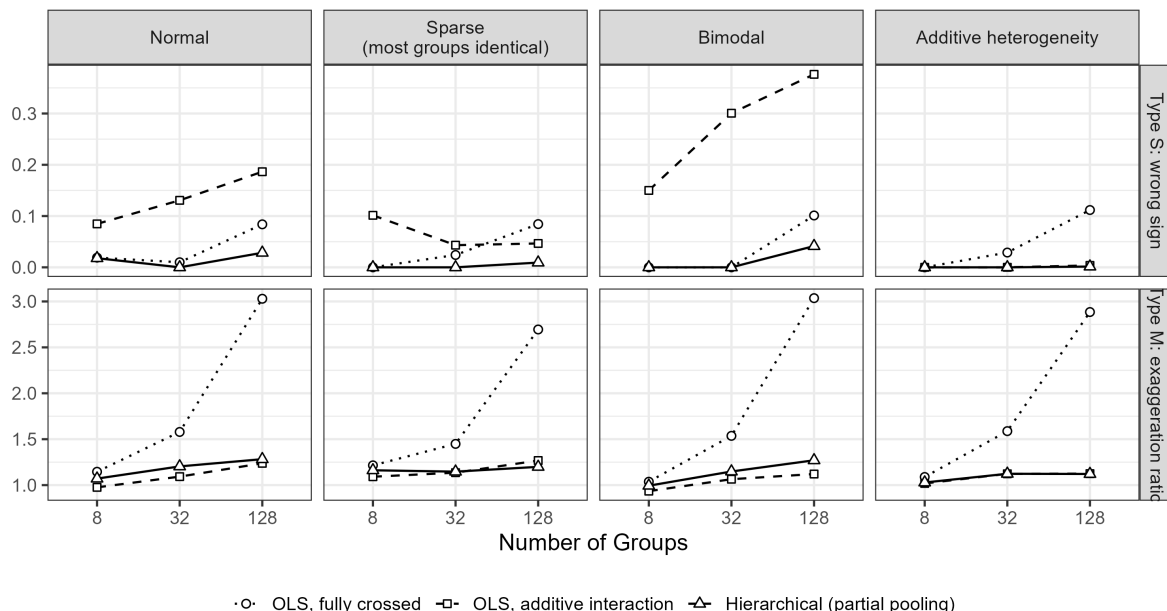


Figure 4. Type M and S error rates.

Fully crossed OLS has more problems of effect magnitude, as statistically significant effects are often large overestimates. With small groups and substantial noise, the exaggeration ratio approaches 3. Additive OLS and hierarchical models are less likely to exaggerate.

The simulation therefore shows the virtues of hierarchical modeling in three steps. In the aggregate, hierarchical models outperform the most common OLS specifications because their estimates are less variable, although slightly more biased due to partial pooling. This reduces the rate of misleading conclusions about the sign and magnitude of non-zero effects.

3 Reanalysis: Barber and Pope 2019

To further illustrate how this hierarchical approach works in practice, I reanalyze a study by Barber and Pope (2019). This clever paper attempts to disentangle ideology and party affiliation with an experiment that attributes liberal or conservative cues to Donald Trump.

They show that liberal cues from Trump move Republican attitudes in a liberal direction, while conservative cues move the same attitudes the other way.

After showing the average effects, Barber and Pope (BP henceforth) analyze how knowledge, party attachment, Trump approval and ideology modify the impact of these cues. They do this with four distinct models, one for each modifier. I unify these tests in a single hierarchical framework, which is crucial because as BP acknowledge, these four modifiers are highly correlated (Barber and Pope, 2019, pg. 50). As an additional comparison, I also fit an interaction model that enters each modifier additively. Details of the grouping decisions, diagnostics of the Bayesian model, and example code are available in the appendix.

3.1 Results

Figure 5 plots marginal effects under three different approaches: the original BP separate models, one unified interaction, and the hierarchical model. In this figure, positive values imply that a cue moves policy in a more liberal direction. Negative values imply more conservative preferences.

While BP’s four models find that each modifier changes the impact of Trump cues in a statistically significant way, unifying the estimates under a single model yields quite different results. Both the unified interaction model and the hierarchical model give less precise estimates, and usually less extreme point estimates as well. The hierarchical estimates are the most uncertain, but that additional variance is a true reflection of the correlations in these modifiers.

Most of the estimates retain the same sign, at least, or move from positive or negative to more negligible. This suggests that the direction of the relationship in the aggregate makes sense, but that this design cannot sort out their independent effect. The strong correlations between party identification, Trump approval and ideology make it hard to identify unique moderating effects.¹⁵ Political knowledge is more orthogonal, and this makes the estimates

¹⁵Given the correlations in these four modifiers, another approach might have been to use a factor analysis or

Interaction Effects: Barber and Pope 2019

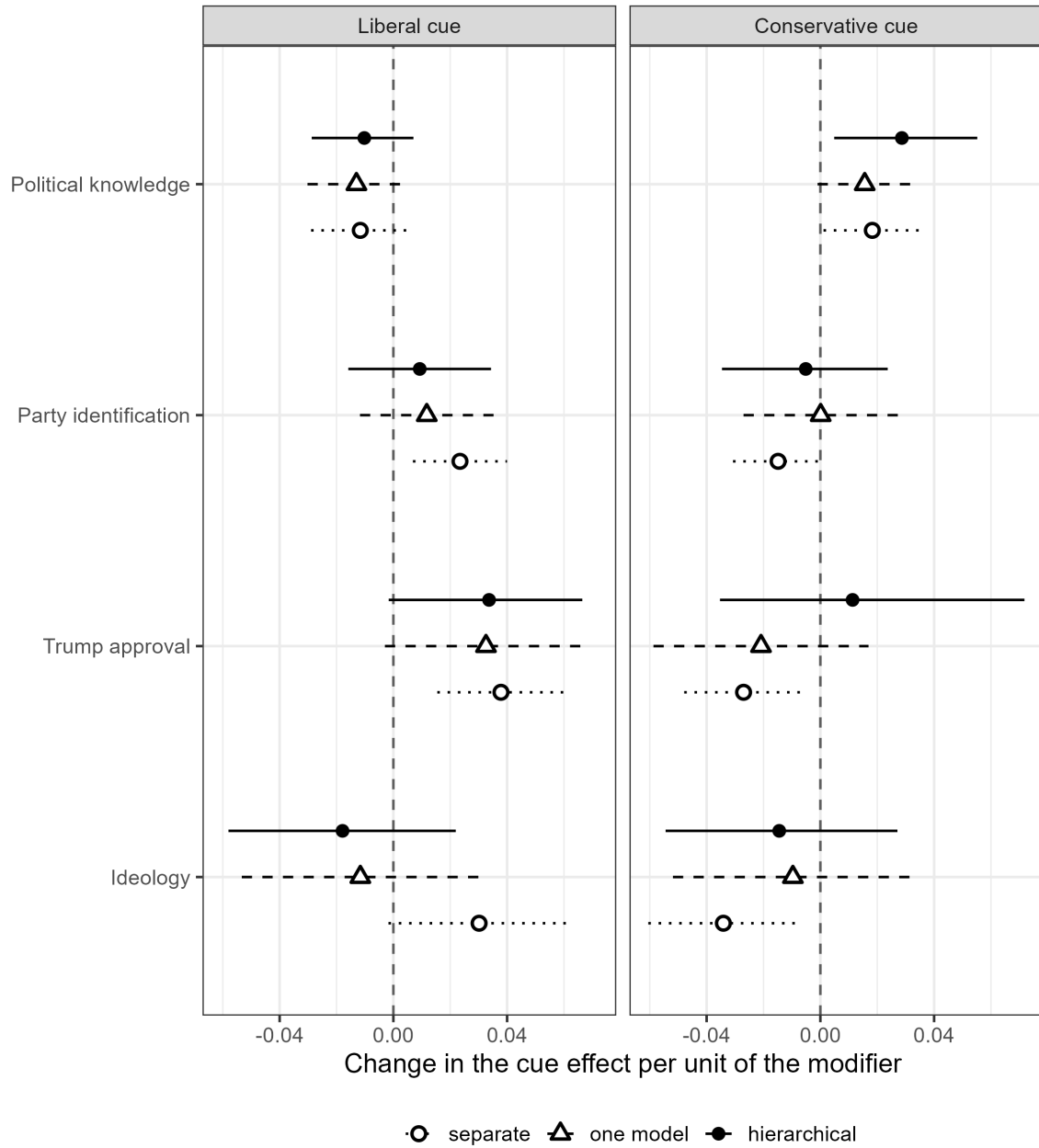


Figure 5. Estimates of the marginal effect of liberal and conservative Trump cues on policy attitudes. Error bars show the 95% confidence or credible interval.

more precise.

Perhaps as a result, only political knowledge has a clearly non-zero relationship with cue responses, once the modifier models are unified. It also moves in a somewhat unexpected direction, as conservative cues from Trump move attitudes in a more liberal direction as political knowledge increases. This is also what BP find, and they attribute it to greater party loyalty among less knowledgeable Republican voters.

The hierarchical model also provides some information about residual effect variation that the modifiers do not capture, akin to the residual in a regression model. For conservative cues, the standard deviation of the treatment effects is .1, while it is .03 for liberal cues. The four modifiers account for most of the variation in responses to liberal cues. The hierarchical model picks up much more of the variation the modifiers miss in the conservative cues analysis. Thus, the hierarchical model gives a more accurate picture of the uncertainty and explanatory power of the modifiers.

These results show some potential uses of the hierarchical approach to heterogeneous effects. Unifying disparate interactions in a single model is wise, and in doing so, scholars should also regularize those estimates. In this case, regularization makes it very clear that correlations between three of the modifiers make clear conclusions about each particular modifier unsupported.

4 Conclusion

As conditional theories proliferate in social science, many independent variables have multiple potential modifiers. Interaction models cannot easily handle this shift, as adding three or more variables to an interaction reduces the interpretation benefits and fitting separate models for each modifier increases the risk of spurious results. Hierarchical modeling retains the clear other model to capture the common concept or two behind party identification, approval and ideology.

interpretation of interactions in a single model that regularizes estimates.

I have therefore explained how to use hierarchical models to estimate heterogeneous effects. The first step is to identify sources of heterogeneity, usually unit characteristics or other treatments, and set corresponding groups. Then, fit two regressions, one that models the relationship between a treatment and the outcome, and another that models the treatment effect with the modifiers and varying intercepts by group.

Explicitly modeling how different groups respond to an independent variable can help test arguments and inform policy. It also details what drives variation in an effect and how much an effect varies. By using partial pooling of different groups, hierarchical modeling is less vulnerable to spurious inferences about subgroups.

The main downside of this model is that as a researcher adds more groups and group size shrinks, partial pooling becomes less useful. Small groups lack sufficient data to offset the prior, and are thus pooled strongly to the overall mean. If all the groups are small, this reduces the estimated variance in effect sizes so much that in extreme cases, it will collapse to basically zero. At this point, the systematic component in the regression will dominate and produce conclusions that are closer to interaction models.

Hierarchical modeling provides an intermediate approach between interactions and machine learning algorithms. For interactions with one or perhaps two modifiers, relying on simple interaction tools is best. Machine learning is best for discovery of complex heterogeneity, especially in large data. When there are two or more theoretically informed modifiers and researchers want to test their relative weight, hierarchical modeling allows flexible and interpretable estimation of effect variation.

Future work can build on this approach in other ways. Gaussian process regression is another flexible estimation tool with similar benefits that could extend to similar problems (Prati et al., 2023). Considering temporal, spatial and spatiotemporal autocorrelation in both equations is another worthwhile task. Another interesting application might be pooling experimen-

tal data and using the hierarchical structure to compare different manipulations in experiments with similar structure and smaller modifications. These tasks would further expand the benefits of hierarchical modeling in incorporating multiple conditional arguments into a single estimation framework.¹⁶

¹⁶AI Disclosure: I used Claude Code to write and refine the simulation and analysis code, including the analysis of prior publications encapsulated in figure 1. The classification itself also uses an open-weight language model (gpt-oss-120b). I also checked all the code for bugs or problems with Claude Code.

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Appendix: Using Hierarchical Models to Estimate Heterogeneous Effects

Contents

1	Replication Analysis Details	3
1.1	Classification	3
1.2	Validation	3
2	Simulation Data Generating Process	4
2.1	Common Structure	4
2.2	The Eight Data Generating Processes	5
2.3	Replication and Estimation	6
3	Comparison with Regularized Machine Learning	7
4	Binning Continuous Modifiers	9
4.1	Smoothly Varying Effects	9
4.2	Bin Count and Cell Size	9
4.3	Cut Points	11
4.4	Ordered Bins	11
4.5	Practical Guidance	11
5	Barber and Pope Reanalysis	13
5.1	Data and Groups	13
5.2	Model and Code	13
5.3	Priors	14
5.4	Standard Errors in the Published Models	15
5.5	Joint Tests	15
5.6	Group Variation and Robustness	15
5.7	Sampler Diagnostics	16
5.8	Out-of-Sample Accuracy	16

6	Tomz and Weeks Reanalysis	17
6.1	Model and Code	17
6.2	Priors	18
6.3	Specification and Pooling	18
6.4	Sensitivity to the Prior, Likelihood, and Grouping	20
6.5	Sampler Diagnostics	21

1 Replication Analysis Details

To produce Figure 1, I downloaded every replication archive on the Harvard Dataverse for papers in the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics*. Coverage runs from 2012 to 2024 and includes 1,830 archives, as deposits for 2025 and 2026 remain incomplete. Each paper is dated by its year of publication, taken from the Dataverse citation string. I use date of publication because journals adopted deposit mandates around 2015 (Rainey et al., 2025), so older articles were archived retroactively.

Within each archive I read every file of author-written analysis code, meaning scripts in R, Stata, Python, Stan, and JAGS. This excludes `.ado` files, which are third-party Stata packages rather than the author’s analysis. Their help text contains the same syntax the analysis looks for. The 1,830 archives contain 12,179 such files.

1.1 Classification

Rather than match patterns, I gave each file in full to an open-weight language model and asked whether the analysis estimates an effect that varies across units, subgroups, or contexts. The model returns a structured judgment, the methods it identifies, and verbatim lines of code as evidence. The model splits files longer than the context window into overlapping chunks and judges them in turn.

Pattern matching was inadequate here, and in a way that affects the trend as well as the level. Regular expressions key on the syntax of an interaction, and that syntax changed over the period. Stata’s factor-variable notation arrived in version 11 and displaced hand-constructed product variables only gradually, so a script from 2012 is far more likely to write `gen treatXpost = treat*post` and enter the result as an ordinary regressor. Scoring the same with pattern matching and the language model, the two approaches agree almost exactly on recent papers and diverge sharply on old ones. Over 2020 to 2024 both flag 67% of papers, while over 2012 to 2015 pattern matching flags 35% against the classifier’s 58%.

I use `openai/gpt-oss-120b`, served through DeepInfra at temperature zero. The prompt, the full text of every request, and every raw model response are archived in the replication materials, so the classification can be verified without re-running inference.

1.2 Validation

I hand-adjudicated a stratified random sample of 100 file-chunks, drawn blind. I shuffled the sample and hid the model’s judgment until each item was answered. Each item displayed the relevant code with four lines of context on either side. Weighting back to the stratum populations, precision for the measure reported in Figure 1 is above 90%.

Two limits apply. First, recall is not identified by this design. Estimating recall requires knowing how often the classifier misses a heterogeneous effect entirely. Over half the corpus falls in the stratum where nothing was flagged, so there is nothing to check without reading whole files. Figure 1 is therefore a lower bound. Second, I do not count subgroup analysis

conducted by splitting the sample and fitting the same model separately, though that estimator is algebraically identical to a fully crossed interaction for the group-specific point estimates. I exclude it because it is the least reliably measured category. Agreement between two independently prompted models on subgroup splitting was $\kappa = 0.49$, against 0.92 for the measure reported here.

2 Simulation Data Generating Process

This section summarizes the data generating processes for the simulations in the main text. All of them share one structure and differ only in two places: the probabilities used to assign the grouping variables, which control how uneven the groups are, and the distribution from which the true group treatment effects are drawn. Every condition fixes the sample size at $n = 2000$ and varies the number of groups.

2.1 Common Structure

I generate k binary grouping variables, where $k \in \{3, 4, 5, 6, 7\}$:

$$x_{ij} \sim \text{Bernoulli}(p_j) \quad \text{for } j = 1, \dots, k$$

The simulation defines the groups by unique combinations of these variables, producing 2^k groups. With three modifiers, there are 8 groups, while 7 modifiers produces 128 groups. Most processes set $p_j = 0.5$ for all j , which makes the groups equal in expectation. The two unbalanced processes instead space the probabilities evenly over $[0.35, 0.65]$. With $k = 7$ this is already severe, giving expected cell sizes that run from roughly one observation to roughly one hundred.

The treatment indicator is randomly assigned, also with a bernoulli draw:

$$T_i \sim \text{Bernoulli}(0.5)$$

Three control variables predict the baseline outcome but do not modify the treatment effect:

$$\begin{aligned} z_{i1} &\sim N(0, 1) \\ z_{i2} &\sim \text{Bernoulli}(0.5) \\ z_{i3} &\sim N(0, 1) \end{aligned}$$

The outcome is then generated as:

$$\begin{aligned} y_i &= \beta_1 z_{i1} + \beta_2 z_{i2} + \beta_3 z_{i3} + \tau_{g[i]} T_i + \epsilon_i \\ \epsilon_i &\sim N(0, \sigma_y) \quad \text{where } \sigma_y = 1 \end{aligned}$$

I also fix the control coefficients $\beta = (0.4, -0.3, 0.25)$. Throughout, $\mu_\tau = 0.2$ and $\sigma_\tau = 0.3$, and each process is scaled so that the marginal standard deviation of the group effects stays near σ_τ . Holding that constant keeps root mean squared error comparable across processes.

2.2 The Eight Data Generating Processes

Table 1 lists them. Let $\mathbf{Z}_g$ collect the modifier values for group g , and let

$$\gamma = (0.20, -0.15, 0.12, -0.10, 0.08, -0.06, 0.05)$$

truncated to the first k entries. Where a systematic component $\mathbf{Z}_g\gamma^*$ appears it is centered and rescaled to the standard deviation shown. In 1, p_j gives bernoulli probabilities on grouping variable values.

Process	p_j	Group treatment effects
Normal	0.5	$\tau_g \sim N(\mu_\tau, \sigma_\tau)$
Systematic + deviation	0.5	$\tau_g = \mu_\tau + \mathbf{Z}_g\gamma^* + \nu_g$, with $\mathbf{Z}_g\gamma^*$ scaled to SD $\sigma_\tau/\sqrt{2}$ and $\nu_g \sim N(0, \sigma_\tau/\sqrt{2})$
Sparse	0.5	$\tau_g = \mu_\tau + d_g\nu_g$, where $d_g \sim \text{Bernoulli}(0.15)$ and $\nu_g \sim N(0, \sigma_\tau/\sqrt{0.15})$: most groups share one common effect and a minority genuinely differ
Bimodal	0.5	$\tau_g = \mu_\tau \pm 0.3 + \nu_g$, sign drawn Bernoulli(0.5), $\nu_g \sim N(0, 0.1)$: two clusters, with little mass at the grand mean
Student- t	0.5	$\tau_g = \mu_\tau + \sigma_\tau t_3/\sqrt{3}$: heavy tails, so genuine outlier groups exist
Unequal group sizes	[0.35, 0.65]	$\tau_g \sim N(\mu_\tau, \sigma_\tau)$: effects as in the normal case, but groups differ enormously in size
Unequal sizes, effects correlated with size	[0.35, 0.65]	$\tau_g = \mu_\tau + \sigma_\tau(0.8z_g + \sqrt{1 - 0.8^2}\nu_g)$, where z_g is the standardized log expected size of group g : the groups with the most data pull the grand mean toward themselves
Exactly additive	0.5	$\tau_g = \mu_\tau + \mathbf{Z}_g\gamma^*$ scaled to SD σ_τ , with <i>no</i> idiosyncratic component

Table 1. Data generating processes. All share the common structure above and differ only in the assignment probabilities and the distribution of the group treatment effects.

The last process needs some comment, because it is the one designed against the hierarchical model. In every other process the group effects contain an idiosyncratic component that an additive interaction model cannot represent by construction, which is a comparison the hierarchical model cannot lose. Under the exactly additive process the additive model is correctly specified, and the hierarchical model estimates a group variance that is genuinely zero. This is why the hierarchical model is roughly three percent of root mean squared error worse in this process.

2.3 Replication and Estimation

Each condition uses 200 Monte Carlo replications, and the shaded bands in the figures give 95% Monte Carlo intervals computed from the spread of per-replication root mean squared error. Seeds are a deterministic function of the process, the number of modifiers, and the replication index, so any single condition can be reproduced in isolation.

Accuracy and errors of inference come from different estimators. Root mean squared error requires no standard error, so it is computed over all 200 replications using restricted maximum likelihood, which is fast enough to make that many replications affordable. Restricted maximum likelihood corresponds to no prior at all and therefore regularizes less than any half- t or half-normal prior, so it is a conservative stand-in. Where the additive model loses to it, the additive model loses to the Bayesian model by more. Sign and exaggeration errors, however, depend on which estimates are declared distinguishable from zero, and the mixed model has no convenient interval for a group treatment effect. The prediction error is $\mathbf{x}'_g(\hat{\beta} - \beta) + (\hat{\theta}_g - \theta_g)$, and the covariance between those two terms is strongly negative, because the varying slope absorbs error in the fixed part. Adding the two variances yields intervals covering 0.99 against a nominal 0.95, while the conditional variance alone covers 0.84. Those quantities are therefore computed from posterior intervals, whose coverage across these processes measures between 0.94 and 0.97. The posterior tier uses 12 replications per condition, fit in brms under the normal, sparse, bimodal, and exactly additive processes at 8, 32, and 128 groups.

The same tier checks sensitivity to the prior on σ_θ . Replacing the brms default, a half- $t(3, 0, 2.5)$, with a half-normal(0, 1) changed root mean squared error by at most 0.001 under the normal, systematic, sparse, and bimodal processes at 8 and 128 groups. With 2,000 observations the group standard deviation is identified well enough that any weakly informative prior gives the same answer.

3 Comparison with Regularized Machine Learning

While interactions are very common, researchers have other tools with which to regularize effect estimates. This section compares the hierarchical model with three of them with a LASSO applied to the saturated interaction expansion (Ratkovic and Tingley, 2017; Blackwell and Olson, 2022), a causal forest (Wager and Athey, 2018), and Bayesian additive regression trees (Dorie et al., 2022).

All four estimators recover the same quantity, the treatment effect in each of the 2^k groups, so the comparison is on one estimand. The LASSO is fit over the full interaction basis and penalized by cross-validation. The forest and the trees estimate an effect for each observation, which I average within group. The data generating processes are those in the previous section, and each condition uses 50 Monte Carlo replications rather than the 200 used elsewhere, because a forest or a set of regression trees takes much longer to fit than a mixed model.

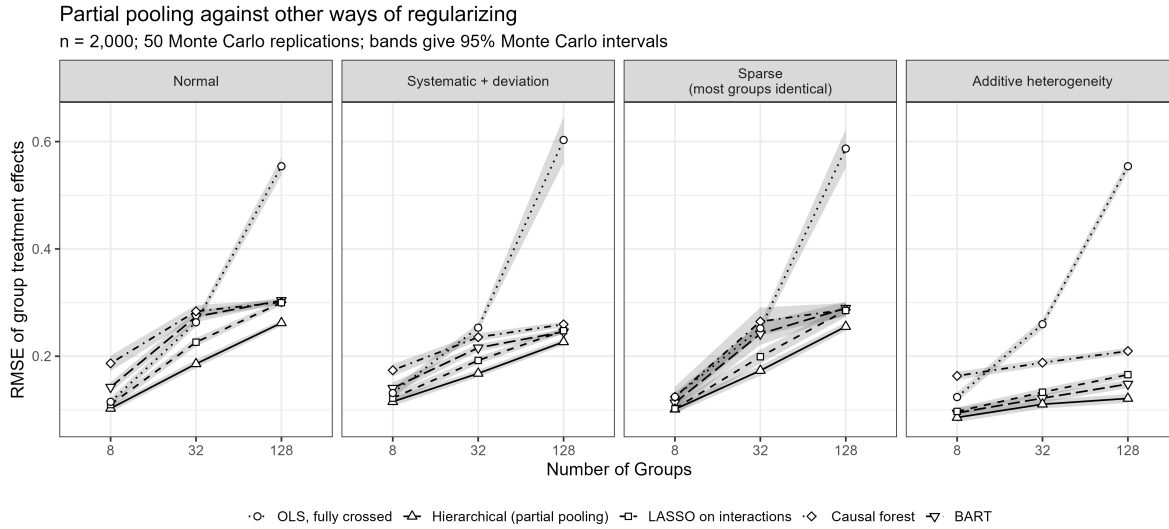


Figure 1. Accuracy of group treatment effect estimates under partial pooling and three regularized alternatives. Bands give 95% Monte Carlo intervals.

Figure 1 shows the result. The three regularized estimators are similar and far more accurate than the fully crossed model. With 128 groups under the normal process, root mean squared error is 0.30 for the LASSO, 0.30 for the forest and 0.30 for the trees, against 0.55 for saturated ordinary least squares.

The hierarchical model is more accurate than all three in every condition, at 0.26 under the same setting. Paired within replication, the differences run from 0.007 to 0.098 and exclude zero in all but three of the forty-eight comparisons. The LASSO is the closest competitor because penalizing the interaction basis and pooling group deviations do similar things. LASSO shrinks departures to zero and the hierarchical model shrinks them toward the mean. The causal forest is consistently the least accurate of the three, likely due to a smaller sample.

There are two important qualifications. First, the hierarchical advantage falls under sparse

heterogeneity and few groups. With eight groups under the sparse process, the difference from Bayesian additive regression trees is 0.012 with an interval spanning zero, and the hierarchical model is more accurate in only 48% of replications.

Second, and more important, this comparison is run on terms that suit the hierarchical model. The modifiers are known in advance, binary, and the grouping is given. Forests and regression trees are built to discover heterogeneity in high dimensional and continuous covariates. Therefore, when researchers already know which modifiers matter and want effects for interpretable groups, partial pooling estimates those effects more accurately than a generic regularizer. Where the modifiers are unknown, machine learning tools are much better.

4 Binning Continuous Modifiers

Grouping observations on a continuous modifier requires choosing how many bins to use and where to place the cuts. Both are researcher degrees of freedom that could hinder replication. I provide some simulation guidance in this section to inform design and pre-registration.

Measuring this requires a different estimand. With binary modifiers there is a finite set of groups, each with a true effect, and accuracy can be measured against those. A continuous modifier has no groups until the analyst makes them, and every binning scheme implies a different set of true group effects. As a result, I cannot compare true to estimated without a circular analysis. Accuracy is therefore measured against the true conditional effect at each respondent, $\tau(x_i)$, which does not depend on how the bins are drawn.

The simulation draws $x \sim \text{Uniform}(0, 1)$ with $n = 2000$, and generates the outcome as in the previous sections. I use three shapes for $\tau(x)$. These are linear, a logistic curve, and a step function that jumps at $x = 0.5$. Cuts are placed at equal widths, at quantiles of x , or at quantiles of the true effect. That last rule is infeasible in practice, and I include it only as an upper bound on what better guidance could achieve.

I then compare five different approaches, including saturated least squares on the bins, partial pooling across bins treated as exchangeable, partial pooling that respects the ordering of the bins through a penalized spline on the bin index, and two estimators that never bin at all, one interacting the treatment with x linearly and one fitting a smooth varying slope. Each condition uses 100 Monte Carlo replications.

4.1 *Smoothly Varying Effects*

When $\tau(x)$ is linear or logistic, an estimator that keeps x continuous is more accurate than any binning of it, at any number of bins. Root mean squared error for the smooth varying slope is 0.064 under the linear process and 0.075 under the logistic one. The best binned estimator reaches 0.069 and 0.078 respectively, and only at eight bins. The gap is not large, but it never closes because when binning replaces a smooth function with a step function, it becomes less accurate.

This is a scope condition. Grouping is worth its cost when the modifiers are genuinely categorical, or when researchers want effects for interpretable groups instead of a curve. When the modifier is continuous and the grouping is not itself of interest, a varying slope on a smooth term is the better tool, it is just harder to integrate with discrete modifiers.

4.2 *Bin Count and Cell Size*

Comparing the two binned estimators is more useful than comparing them to the continuous ones, because it isolates what pooling does. Under the linear process with quantile cuts, saturated least squares is most accurate at four bins (0.095) and then deteriorates, reaching 0.124 at eight bins as the cells empty out. The hierarchical model improves monotonically over the same range, from 0.109 at two bins to 0.069 at eight.

What binning a continuous modifier costs

$n = 2,000$; 100 Monte Carlo replications. The smooth varying slope never bins and is repeated across bin counts as a benchmark.

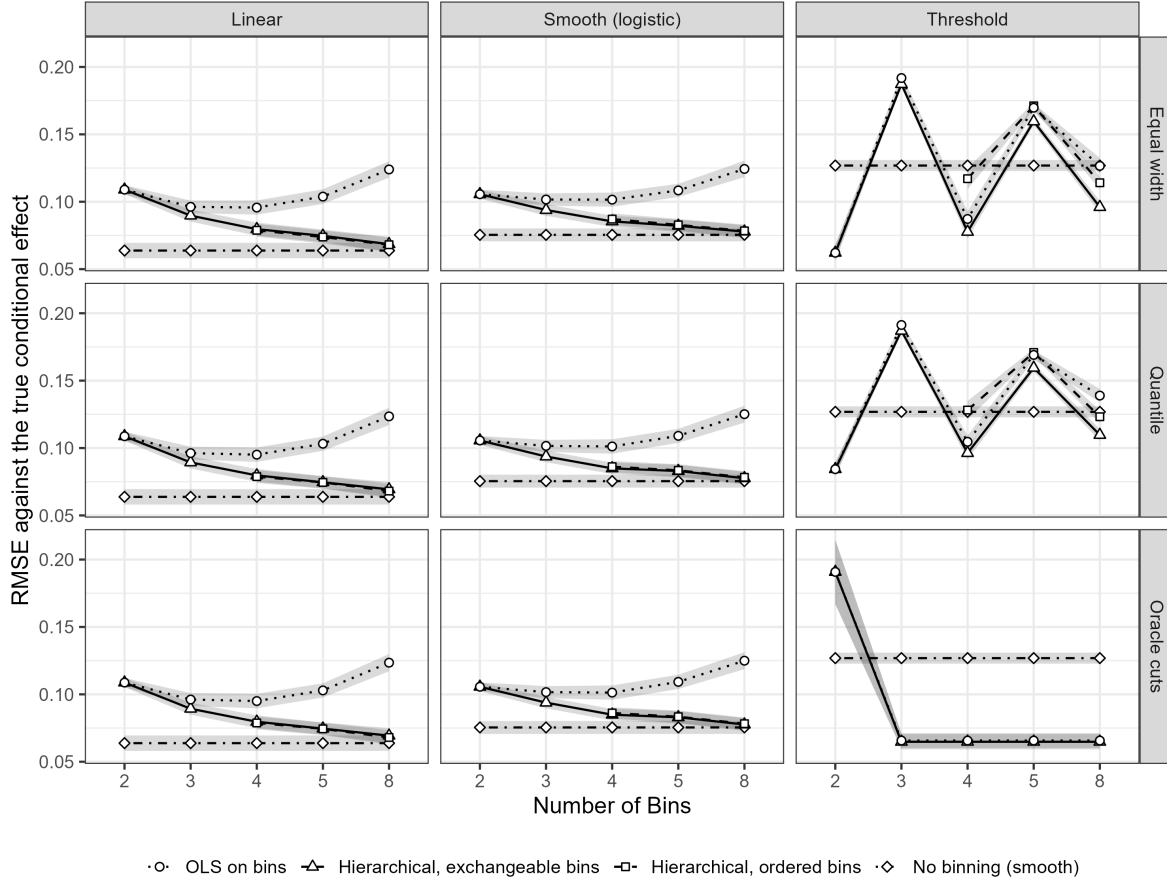


Figure 2. Accuracy against the true conditional effect under different binning schemes. The estimators that never bin do not vary with the number of bins and are repeated across each panel as a benchmark.

Finer bins reduce the approximation error and shrink the cells, and which of those dominates depends on whether the estimator can handle cells with little data. Least squares breaks down with many small cells, so it needs coarse bins that misrepresent the function. Partial pooling can handle finer bins, but inferences about individual bins should be treated with caution because those bins will be regularized..

4.3 *Cut Points*

The threshold process raises the largest concern about researcher degrees of freedom. The true effect jumps at $x = 0.5$, so a cut point at 0.5 represents it perfectly and any other placement cannot. With equal-width cuts the accuracy of the hierarchical model oscillates with the number of bins: 0.062 at two bins, 0.187 at three, 0.078 at four, 0.160 at five, and 0.096 at eight. Even numbers of bins place a boundary at 0.5 and odd numbers do not.

Moving from two bins to three, a change most analysts would regard as innocuous, triples the error. While cut points may seem rare, this is a clear argument for pre-registering a grouping scheme and reporting results under more than one.

4.4 *Ordered Bins*

Treating the bins as exchangeable discards information. Bins 1 and 5 are pulled toward the grand mean as strongly as bins 1 and 2 are pulled toward each other, when adjacent bins should be more alike. A smoothing prior over the ordered bin index is the obvious remedy, but it does not work.

Under the linear and logistic processes the difference between the exchangeable and ordered versions is at most 0.002 in either direction, which is negligible. Under the threshold process the ordered version is clearly worse, by 0.011 to 0.039 depending on the number of bins and the cut rule. The reason is the same one that makes the threshold case hard. Smoothing across bins assumes that the effect varies gradually, and when the truth is a discontinuity that assumption is wrong in the place it matters most.

When a modifier is expected to shift the effect gradually, smoothing across ordered bins is defensible. Where a discontinuity is plausible, exchangeable pooling is the safer default, because it imposes no belief about which bins are similar.

4.5 *Practical Guidance*

These results translate into six recommendations.

1. *Decide whether the groups are of interest.* If the goal is how an effect changes along a continuous modifier, do not bin. Fit a smooth varying slope instead, such as $s(x, by = \text{treat})$ in brms, or a Gaussian process (Prati et al., 2023). Bin only when effects for interpretable groups are the quantity of interest, or when a continuous modifier must be combined with categorical ones.

2. *Keep the trend in the systematic component and model deviation with bins.* This leaves partial pooling to regularize some levels.
3. *Use more bins than least squares could support.* Least squares was most accurate at four bins and deteriorated beyond that, while partial pooling kept improving through eight. Before fitting, count the cells implied by crossing every grouping variable and check the smallest cell in each treatment arm, because crossing multiplies bins quickly.
4. *Put cuts where theory or institutions put a threshold, or use quantiles.* Adding bins does not help when a boundary is in the wrong spot. Quantile cuts keep cells balanced when the modifier is skewed, where equal-width cuts leave the tail bins nearly empty.
5. *Pool bins as exchangeable unless the effect should change gradually.* Smoothing across ordered bins gained nothing where it was correct and cost accuracy where the effect jumped.
6. *Pre-register the number of bins and the cut rule, and report at least one alternative.* As with other issues, pre-registration can help express prior beliefs about these functions. Moreover, if pre-registered models do not fit well, that is informative, because model misspecification often complicates model fitting.

5 Barber and Pope Reanalysis

5.1 Data and Groups

Barber and Pope (2019) randomly assigned respondents to one of four conditions before asking their positions on ten policies. In two, Trump was described as taking the liberal or the conservative position on each policy. In the third, Republican leaders in Congress took a position, and the fourth gave no cue. Their analysis of heterogeneity compares the two Trump cues with the no-cue condition, and so does this reanalysis, which leaves out the party-leader condition. Each respondent answered all ten items under a single condition. After dropping respondents who answered “not sure” to the party identification question or skipped the Trump approval question, the data contain 7,173 item responses from 845 respondents. That is the estimation sample of their Table 1, where the same respondents drop out through missing values on party strength and Trump approval.

The four modifiers are political knowledge, measured as the number of eight factual questions answered correctly, the seven-point party identification scale, Trump approval on a five-point scale, and self-placed ideology on a five-point scale. Knowledge is nearly uncorrelated with the others, while party identification, Trump approval and ideology correlate between 0.62 and 0.68. Their Table 1 uses a folded measure of party strength, estimated among independents and Republicans only. Because the direction of a respondent’s partisanship matters for a Trump cue, the models here use the full seven-point scale instead.

The groups for partial pooling cross party (Democrat, independent, Republican), Trump approval (disapprove, neither, approve), and knowledge (four or fewer correct answers, five or more). That gives 18 groups, which range from 3 to 178 respondents, with a median of 22.

5.2 Model and Code

Each cue has its own heterogeneity equation, in which the four modifiers predict the effect of the cue and a varying intercept for each group is partially pooled. The modifiers also enter the outcome equation, alongside a varying intercept for each group, one for each respondent, since each answered ten items, and one for each policy item. The group intercept in the outcome equation matters. Without it, groups that differ in baseline support can only show that difference through the cue slopes, which inflates the residual variation the model attributes to the cues. In the paper’s two-equation form, the brms formula is:

```
bf(
  Support ~ lamlib * libtrump + lamcon * contrump + ctrl,

  lamlib ~ knowledge + pid7 + trump_approve + ideo5b + (1 | grp),

  lamcon ~ knowledge + pid7 + trump_approve + ideo5b + (1 | grp),

  ctrl ~ knowledge + pid7 + trump_approve + ideo5b + race_white +
```

```

    (1 | grp) + (1 | caseid) + (1 | Question),

nl = TRUE
)

```

Multiplying out the two equations gives a single formula with the same parameters:

```

Support ~ (libtrump + contrump) *
  (knowledge + pid7 + trump_approve + ideo5b) +
  race_white + (1 | caseid) + (1 | Question) +
  (1 + libtrump + contrump || grp)

```

The two are the same model, but with 845 respondent intercepts the non-linear version takes hours to sample and the multiplied-out version minutes, so the results reported here come from the second. Table 2 checks that the two give the same answer. I ran that check on an earlier version of this model, before I added the group intercept to the outcome equation, and the two forms agree to three decimal places.

Coefficient	Two equations	Multiplied out
Liberal cue × knowledge	−0.014	−0.014
Liberal cue × party identification	0.014	0.013
Liberal cue × Trump approval	0.031	0.032
Liberal cue × ideology	−0.001	−0.001
Conservative cue × knowledge	0.026	0.026
Conservative cue × party identification	0.002	0.002
Conservative cue × Trump approval	0.010	0.009
Conservative cue × ideology	0.004	0.004
Group SD, liberal cue	0.097	0.096
Group SD, conservative cue	0.097	0.095

Table 2. *Posterior means from the two parameterizations of the Barber and Pope model, fit without the group intercept in the outcome equation. Group standard deviations are posterior medians.*

5.3 Priors

All coefficients receive $N(0, 1)$ priors, which are weakly informative on the scale of a binary outcome. The standard deviations of the group, respondent, and item intercepts, and the residual standard deviation, take the brms default, a half- $t(3, 0, 2.5)$.

5.4 Standard Errors in the Published Models

Because each respondent answers ten items, the least squares models in the main text cluster standard errors by respondent. The published models do not. Table 3 reproduces their Table 1 and reports both. Clustering matters for the liberal cue, where three of the four interactions are no longer significant at the .05 level. All four conservative-cue interactions remain significant. The comparisons in the main text use clustered standard errors for every least squares model, so that clustering does not drive any difference between them.

Cue	Modifier	Estimate	p , published	p , clustered
Liberal	Knowledge	-0.011	0.041	0.213
Liberal	Party strength	0.028	0.045	0.200
Liberal	Trump approval	0.041	< 0.001	< 0.001
Liberal	Ideology	0.031	0.002	0.058
Conservative	Knowledge	0.019	0.001	0.031
Conservative	Party strength	-0.066	< 0.001	0.004
Conservative	Trump approval	-0.026	< 0.001	0.012
Conservative	Ideology	-0.033	0.001	0.014

Table 3. *Interactions from the four separate models in Table 1 of Barber and Pope (2019), with the published standard errors and with standard errors clustered by respondent.*

5.5 Joint Tests

In the interaction model that enters all four modifiers additively, none of Trump approval, ideology, or party identification is individually significant for either cue. Tested jointly, the three moderate the liberal cue ($\chi^2_3 = 12.98$, $p = 0.005$) but only marginally the conservative cue ($\chi^2_3 = 6.87$, $p = 0.076$). All four modifiers together are significant for both cues ($p = 0.005$ and $p = 0.030$).

5.6 Group Variation and Robustness

The standard deviation of the group-specific cue effects, net of the four modifiers, is 0.027 for the liberal cue (90% interval 0.003 to 0.088) and 0.097 for the conservative cue (0.027 to 0.189). The groups also differ in baseline support for these policies, with a standard deviation of 0.117 (0.079 to 0.173). For the liberal cue, then, the four modifiers account for nearly all of the variation across groups, while for the conservative cue they leave a good deal of it unexplained.

Grouping by party and Trump approval alone, which gives nine groups, yields 0.044 and 0.115 with wider intervals. The subgroup effects from that model correlate 0.997 with the main model's for the liberal cue and 0.975 for the conservative cue, and differ by at most 0.012 and 0.036.

The outcome is binary, and like the original analysis the main model uses a Gaussian likelihood. Under a Bernoulli likelihood, with effects computed as the average change in the predicted probability of support, the subgroup effects correlate 0.997 with the main model’s for the liberal cue and 0.900 for the conservative cue, and differ by at most 0.013 and 0.045.

5.7 *Sampler Diagnostics*

Each model was fit with four chains of 2,000 iterations, the first 1,000 of them warmup, with `adapt_delta` set to 0.95. The coarser model has one divergent transition, and the main model’s largest $\hat{R}$ is 1.012, marginally above the usual threshold of 1.01.

Model	Max $\hat{R}$	Min bulk ESS	Divergent
Gaussian, party $\times$ approval $\times$ knowledge	1.012	522	0
Bernoulli	1.007	894	0
Party $\times$ approval groups	1.006	445	1

Table 4. *Convergence diagnostics for the Barber and Pope models.*

5.8 *Out-of-Sample Accuracy*

To check whether one model estimates the subgroup effects Barber and Pope report more accurately than their separate models, I split the respondents at random into halves, stratified by group and condition, 200 times for each cue. Using one half, each approach estimates the cue’s effect among Democrats, independents and Republicans, among those who disapprove of Trump, neither approve nor disapprove, or approve, among low- and high-knowledge respondents, and among liberals, moderates and conservatives. Those estimates are scored against the raw differences in means in the other half. Because the held-out difference is independent and unbiased, its noise adds the same amount to every approach’s error, and differences in the score are differences in mean squared error. The hierarchical model is fit by restricted maximum likelihood, as in the simulations.

For the liberal cue, most of the gain comes from estimating the modifiers jointly, and pooling costs a little compared to the additive model. For the conservative cue, the additive model is less accurate than the separate models in most replications, and pooling restores the accuracy it loses.

	Liberal cue	Conservative cue
Held-out error, separate models	0.0045	0.0063
Held-out error, one model	0.0036	0.0068
Held-out error, hierarchical	0.0038	0.0062
Hierarchical more accurate than separate	76%	55%
One model more accurate than separate	81%	27%
Hierarchical more accurate than one model	31%	75%

Table 5. *Accuracy of subgroup effect estimates across 200 split-half replications. Percentages give the share of replications in which one approach has lower held-out error than another.*

6 Tomz and Weeks Reanalysis

The section provides another demonstration of the hierarchical approach by reanalyzing a study by Tomz and Weeks (2021). Tomz and Weeks (TW hereafter) examine whether the public is more willing to go to war when the beneficiary of that intervention is a U.S. ally. In a factorial experiment with vignettes, they find a 33% average increase in support for military intervention on behalf of allies, compared to non-allies. This is a large and potentially important relationship because the United States has a global network of allies.

Given the size of the main effect, TW’s paper is an ideal scenario for comparing interactions and hierarchical models. Corresponding interaction effects may be large, and their sample size of 1,200 respondents is not unusual in published work. At the same time, TW estimated many interactions to check how other treatments modify the impact of alliances by fully crossing interactions of their experimental treatments.¹ There are 64 unique treatment groups with anywhere from 11 to 32 respondents, so estimates of the impact of alliances in the 32 pairs of alliance treatment and control groups employ at most 54 data points. As such, hierarchical regularization will likely change some conclusions by pooling noisy estimates in small groups. For ease of comparison, I also compare an OLS model with additive instead of fully crossed interactions. As the simulation results indicate, the fully crossed results are far noisier.

6.1 Model and Code

The alliance effect varies with the other four treatments (regime, stakes, costs, and region), which enter the heterogeneity equation additively. The `treat_group` indicator marks the 32 combinations of those treatments and carries the varying intercept that is partially pooled. The same four treatments also enter the outcome equation as main effects, alongside respondent demographics and foreign policy dispositions. Costs are coded as in the original data, where one indicates low costs.

b f (

¹This is one example of fully crossed interactions in applied research.

```

force ~ lambda * alliance + controls,

lambda ~ regime + stakes + costs + region_txt +
  (1 | treat_group),

controls ~ regime + stakes + costs + region_txt +
  white + male + hawk + intl + dem + rep + age + ed4,

nl = TRUE
)

```

The main effects in the outcome equation are necessary, even though every treatment is randomized. Without them, untreated respondents in every condition share one intercept, so each λ_g estimates the difference between allied respondents in condition g and non-allied respondents in all conditions. That difference adds the condition’s baseline support to its alliance effect. Omitting them changes the answer substantially. In the condition with the only negative fully crossed estimate, non-allied respondents support intervention at 0.90 against 0.47 across all conditions, so a model without main effects moves that estimate from -0.15 to 0.39 . Across all 32 conditions, its estimates correlate -0.52 with the additive least squares estimates, and leave-one-out cross-validation prefers the model with main effects by 35.3 points of expected log predictive density (standard error 9.4). The same requirement applies to the Bush and Prather model below.

6.2 Priors

Table 6 lists the priors. The coefficient priors are set in the code, while the two standard deviations take the brms defaults. All are weakly informative on the scale of a binary outcome, and the default on σ_θ is the weakest of the three priors compared below.

Parameter	Prior	Sensitivity check
Heterogeneity equation coefficients	$N(0, 1)$	
Outcome equation coefficients	$N(0, 1)$	
σ (residual SD)	half- $t(3, 0, 2.5)$	
σ_θ (group SD)	half- $t(3, 0, 2.5)$	half- $N(0, 1)$, half- $N(0, 0.5)$

Table 6. Priors in the Tomz and Weeks model.

6.3 Specification and Pooling

I estimate the alliance effect in each of the 32 conditions three ways: with TW’s fully crossed least squares model, with an additive least squares model that has the same modifiers and no pooling, and with the hierarchical model. Figure 3 plots all three.

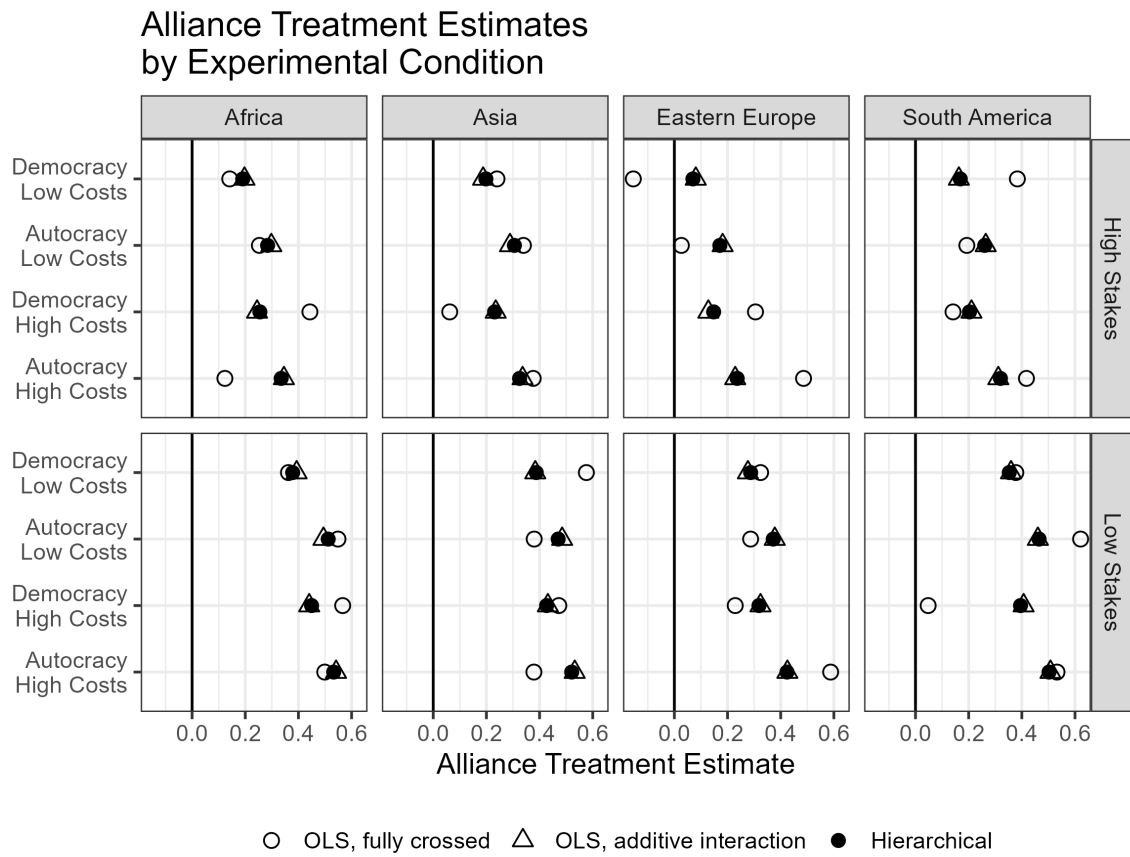


Figure 3. *Estimated impact of alliance in each of the 32 experimental conditions, under fully crossed OLS, additive OLS, and the hierarchical model.*

The additive model separates the two steps that distinguish the hierarchical model from TW's. Moving from the fully crossed to the additive model is specification, and moving from the additive to the hierarchical model is pooling.

Almost all of the change comes from the first step. The standard deviation of the 32 estimates is 0.188 under the fully crossed model, 0.124 under the additive model, and 0.122 under the hierarchical model, and the hierarchical estimates correlate 0.997 with the additive ones. Pooling adds little because the data support little variation beyond the additive structure. The posterior median of σ_θ is 0.031. That estimate comes from the data rather than the prior, since it holds under the weakest prior considered. In this application, then, the hierarchical model estimates that the condition-specific variation is small, where the additive model can only assume it.

The models rarely disagree about effect directions. The fully crossed and hierarchical models assign the same sign to 31 of 32 conditions, and the additive and hierarchical models to all 32. The exception is the only negative fully crossed estimate, -0.154 for a low-cost, high-stakes intervention on behalf of an Eastern European democracy, which the hierarchical model places at 0.070.

The models disagree much more about effect magnitudes. For a high-cost, low-stakes intervention on behalf of a South American democracy, the fully crossed estimate is 0.048 and the hierarchical estimate 0.395.

6.4 *Sensitivity to the Prior, Likelihood, and Grouping*

Replacing the default prior on σ_θ with a half-normal(0, 1) or half-normal(0, 0.5) moves no condition estimate by more than 0.005, and the posterior median of σ_θ stays between 0.031 and 0.032.

A Bernoulli likelihood with a logit link changes the scale on which the modifiers combine, so its effects are computed as the average change in predicted probability within each condition. Those estimates correlate 0.97 with the Gaussian ones and differ by at most 0.10. Every condition has a positive estimate under both likelihoods, the smallest being 0.070 under the Gaussian model and 0.127 under the Bernoulli. The usual weakness of the linear probability model shows up here, as 6% of its expected values for individual respondents fall outside $[0, 1]$. It does not distort the condition estimates, which the Bernoulli model reproduces closely. As a posterior predictive check, the observed mean in each of the 64 alliance-by-condition cells falls inside its 90% posterior predictive interval in 63 cells under the Gaussian model and 62 under the Bernoulli.

Grouping is a researcher choice, so I refit the model with region dropped from the heterogeneity equation, which leaves eight groups instead of 32. The spread of the estimates is unchanged, with a standard deviation of 0.121 against 0.122. Leave-one-out cross-validation cannot distinguish the two groupings. The 32-group model trails by 1.0 points of expected log predictive density, with a standard error of 1.9. The conclusions do not depend on whether region defines the groups.

6.5 Sampler Diagnostics

Every model was fit with four chains of 2,000 iterations, the first 1,000 of them warmup, and `adapt_delta` set to 0.95. Table 7 reports convergence. The coarse model has a single divergent transition, and the others have none.

Model	Max $\hat{R}$	Min bulk ESS	Divergent	σ_θ
Default prior (half- t)	1.005	1,289	0	0.031
Half-normal(0, 1)	1.004	1,523	0	0.031
Half-normal(0, 0.5)	1.003	1,184	0	0.032
Bernoulli likelihood	1.005	873	0	0.282
Coarse grouping (8 groups)	1.003	1,224	1	0.029

Table 7. *Convergence diagnostics for the Tomz and Weeks models. The Bernoulli model's σ_θ is on the logit scale.*

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